

WHY DATA LAKES MATTER: A MUST-KNOW TECHNOLOGY FOR EVERY DATA ENGINEER



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WHAT IS DATA LAKE

A Data Lake is a centralized repository that stores large volumes of data in its original format.

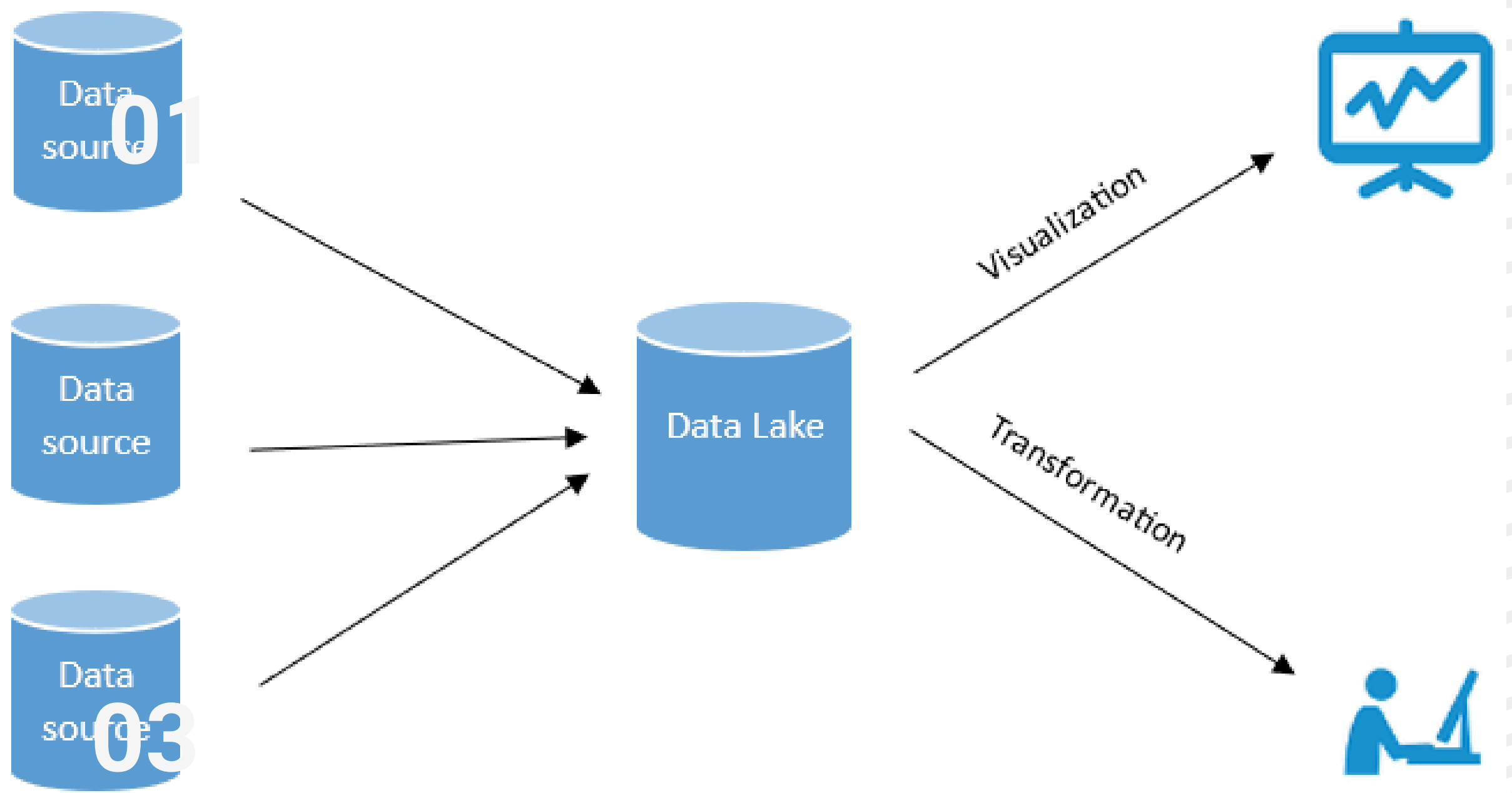
Unlike traditional databases, data is stored first and transformed later when needed.

It Stores

Structured Data (Tables, SQL Databases)

Semi-Structured Data (JSON, XML, Avro)

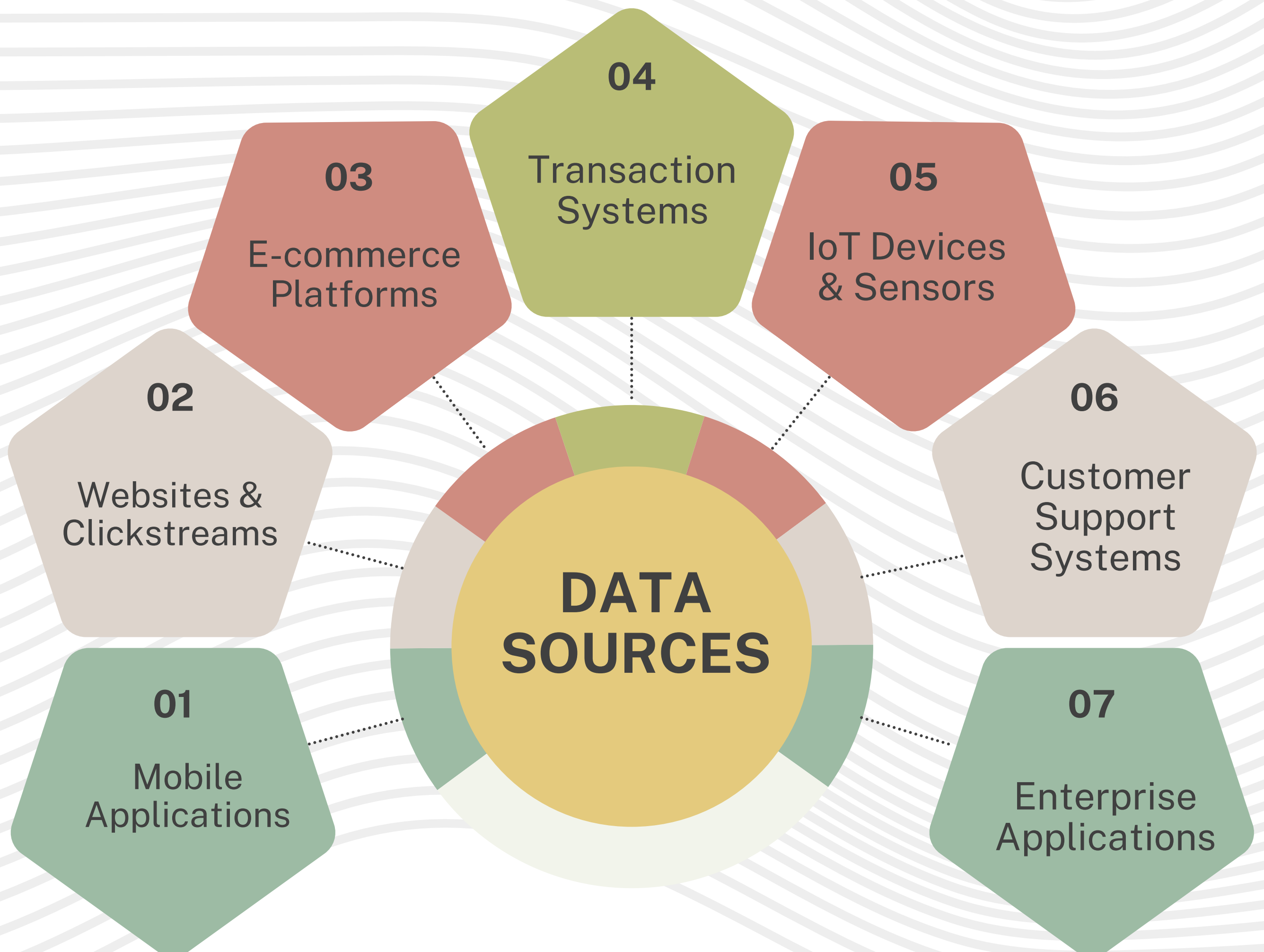
Unstructured Data (Images, Videos, PDFs, Logs)



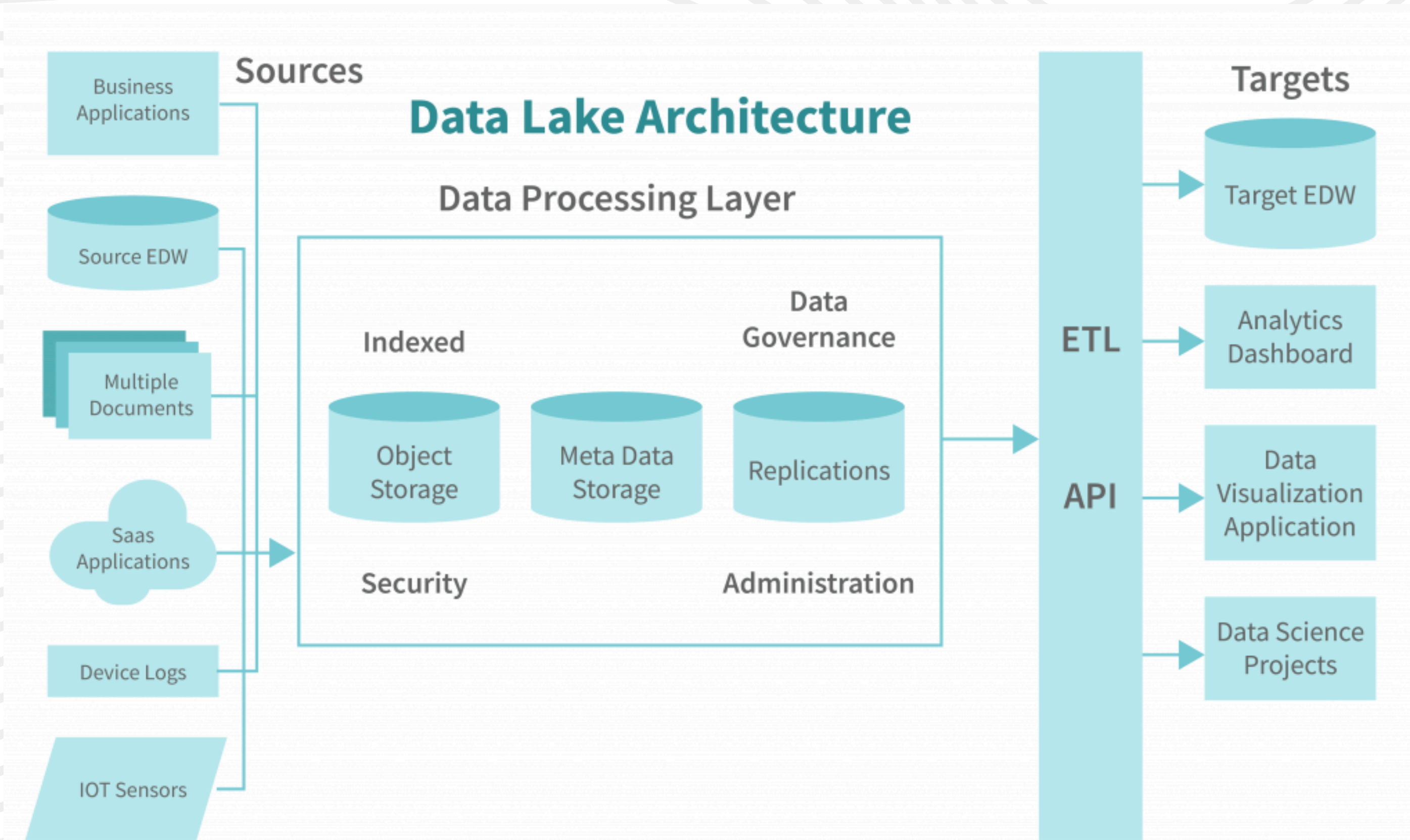
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WHY DO WE NEED DATA LAKES?

Modern organizations generate enormous amounts of data from various sources every second. Traditional warehouses can become expensive and restrictive when handling diverse data formats. Data Lakes solve this problem by providing scalable and cost-effective storage for all data types.



DATA LAKE ARCHITECTURE



CORE CHARACTERISTICS OF A DATA LAKE

01

Schema-on-Read

Data structure is applied when data is queried rather than when it is stored.

05

High Flexibility

Allows teams to explore new use cases without redesigning storage architecture

04

Multi-Framework Support

Works with SQL engines, Spark, Machine Learning tools, and streaming platforms.

02

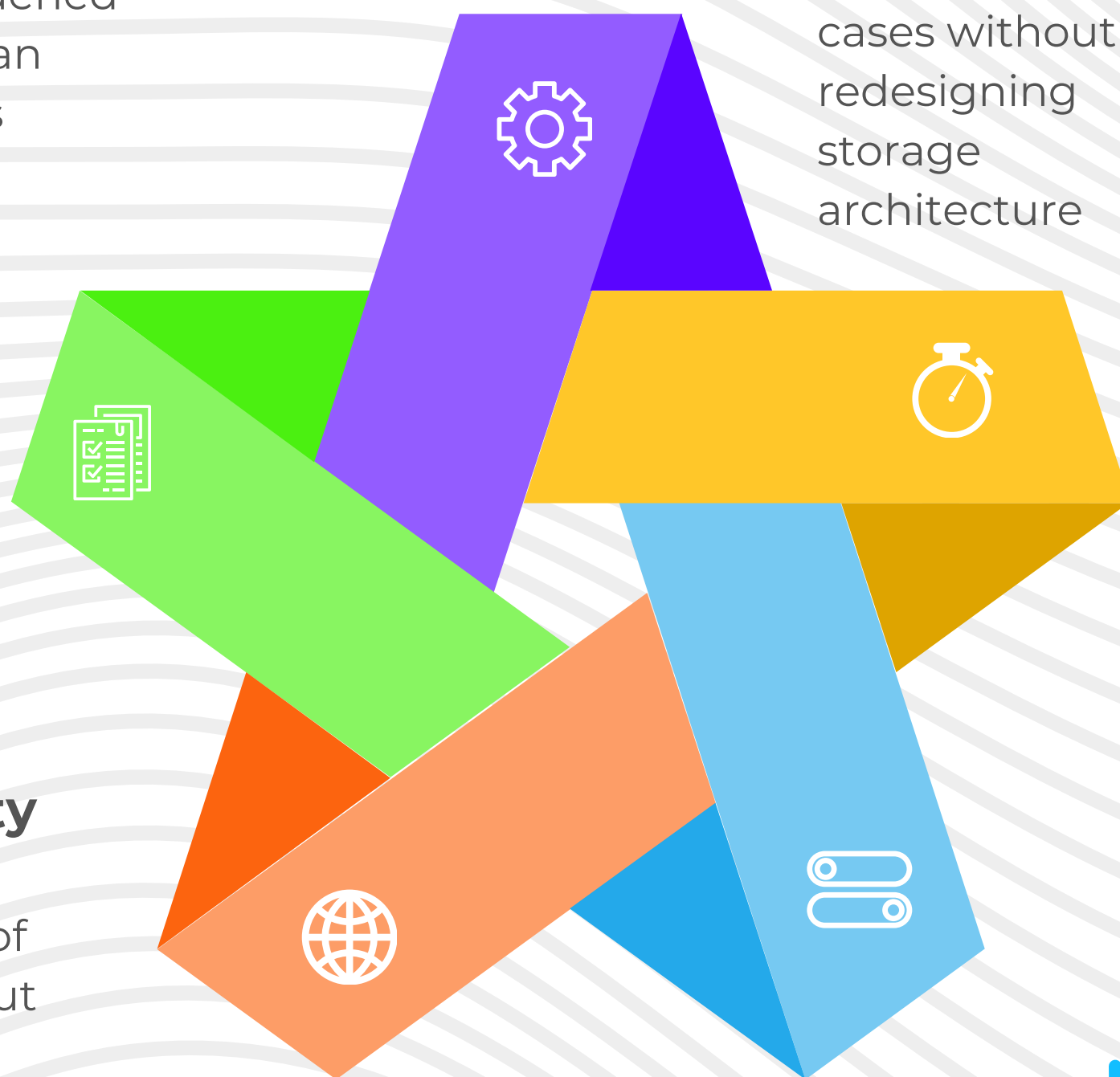
Massive Scalability

Can store petabytes of data without significant infrastructure changes

03

Low-Cost Storage

Uses object storage technologies that are cheaper than traditional database systems.



DATA LAKE ZONES

Bronze Layer (Raw Data)

Stores original source data exactly as received.

Purpose:

- Historical retention
- Auditability
- Reprocessing capability



Silver Layer (Clean Data)

Data is validated, standardized, and enriched.

Purpose:

- Improved quality
- Consistent schemas
- Data cleansing



Gold Layer (Business Data)

Curated datasets ready for reporting and analytics.

Purpose:

- Dashboards
- KPI Tracking
- Machine Learning Features

BENEFITS OF DATA LAKES

Centralized Storage

Consolidates enterprise data into a single platform.

Cost Efficiency

Stores large volumes of data at a lower cost compared to traditional warehouses.

Faster Innovation

Teams can experiment with new datasets without waiting for extensive modeling.

Supports Advanced Analytics

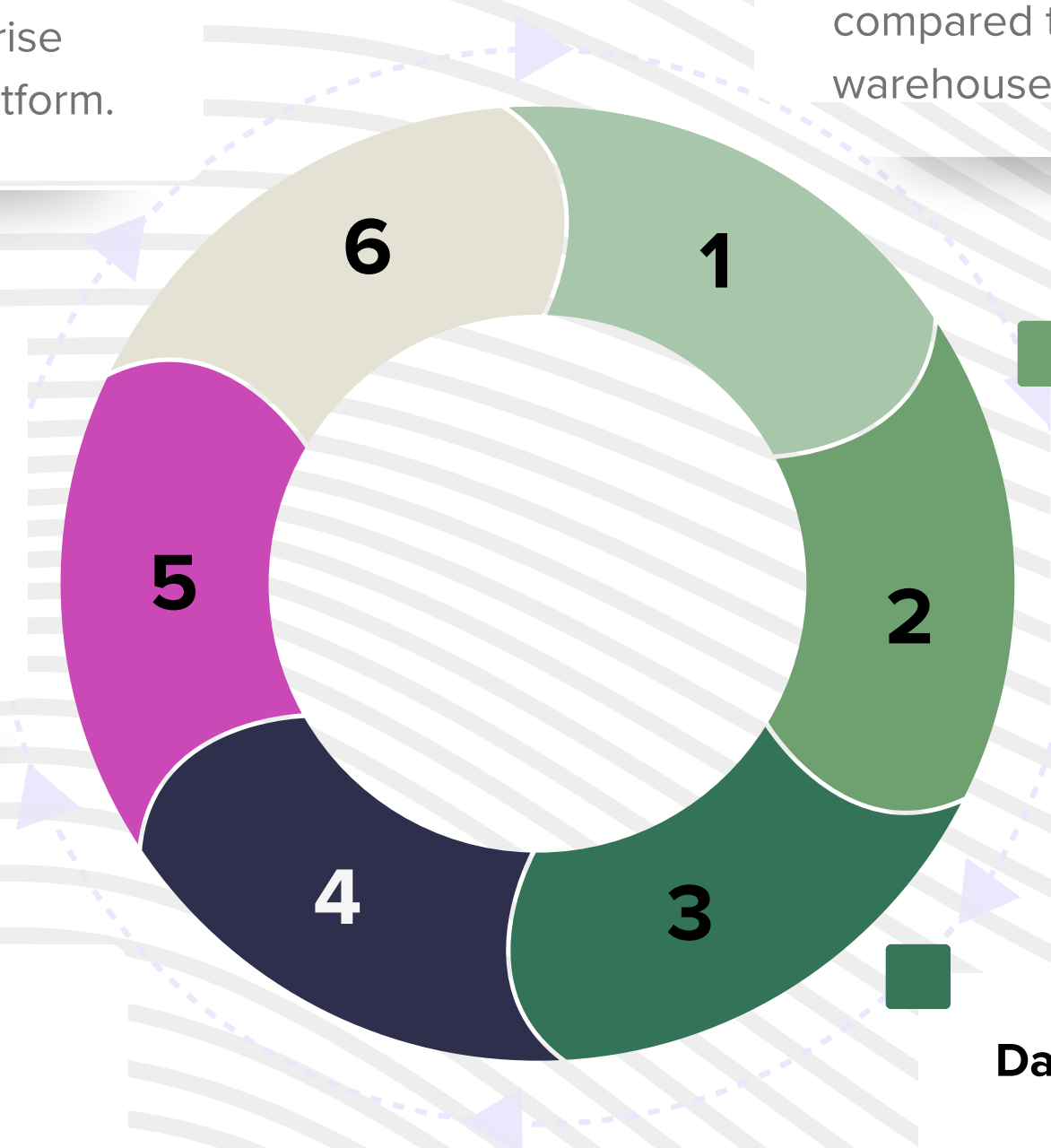
Enables machine learning, predictive analytics, and AI workloads.

Scalability

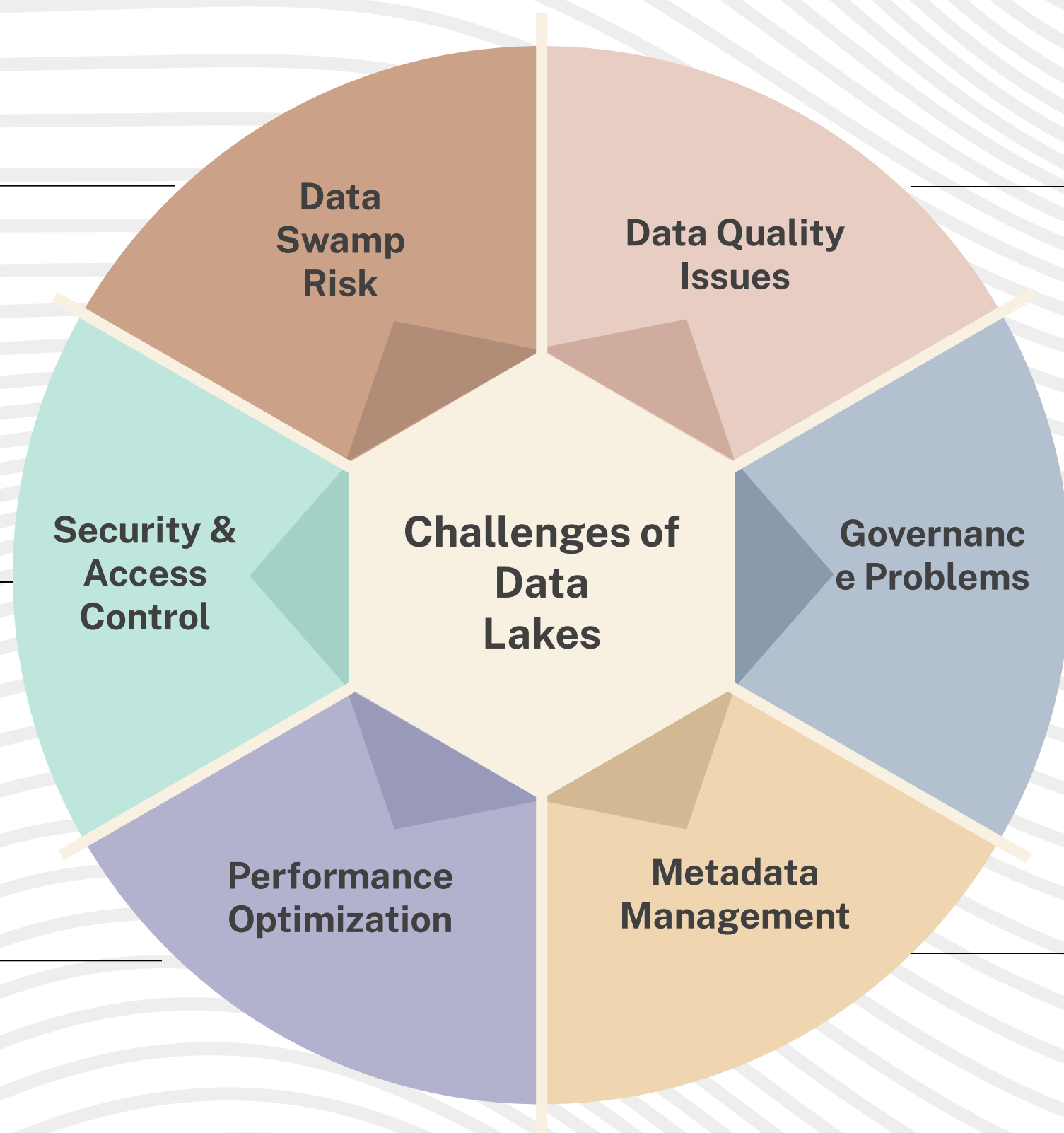
Easily grows from gigabytes to petabytes.

Data Democratization

Makes data accessible to analysts, engineers, scientists, and business users.



CHALLENGES OF DATA LAKES



MODERN DATA LAKES



Amazon S3

Industry-standard object storage service from AWS. Provides virtually unlimited scalability, high durability, and cost-effective storage for Data Lakes.

Azure Data Lake Storage (ADLS)
Microsoft's enterprise-grade storage solution optimized for analytics workloads. Integrates seamlessly with Azure Synapse, Databricks, and Microsoft Fabric.



Azure Data Lake Storage



Google Cloud Storage

Highly scalable object storage platform from Google Cloud. Frequently used for big data analytics, AI/ML workloads, and cloud-native applications.

WHY EVERY DATA ENGINEER SHOULD LEARN DBT

1. Build Modern Data Architectures

Data Lakes are a core component of Data Warehouses, Lakehouses, and cloud-native analytics platforms used by leading organizations.

2. Handle Any Type of Data

Learn how to manage structured, semi-structured, and unstructured data from multiple sources within a single platform.

3. Process Data at Scale

Work with massive datasets using distributed processing frameworks such as Apache Spark, Trino, and Flink.

4. Power AI & Machine Learning

Data Lakes provide the large-scale historical data required for training, testing, and deploying machine learning models.

5. Master Cloud Data Engineering

Most cloud platforms—including AWS, Azure, and Google Cloud—use Data Lakes as the foundation for modern analytics ecosystems.

6. Prepare for the Lakehouse Era

Technologies like Apache Iceberg, Delta Lake, and Apache Hudi are transforming Data Lakes into reliable Lakehouse architectures.

The background of the image consists of numerous thin, light gray lines that flow in a wavy, undulating pattern across the entire frame. These lines create a sense of movement and depth, resembling a stylized topographical map or a liquid surface. The lines are more densely packed in some areas and more spread out in others, contributing to the overall organic feel of the design.

THANK YOU